

# EMERGING TRENDS AND TECHNOLOGIES

## Business, People, and Technology Tomorrow

# STUDENT LEARNING OUTCOMES

1. Describe emerging trends and technologies that will impact the changing Internet.
2. Describe emerging technologies for physiological interaction with technology.
3. Describe emerging trends of Near Field Communication, Bluetooth, Wifi, cell phones, and RFID.
4. Define and describe emerging “pure” technologies that will impact the future.

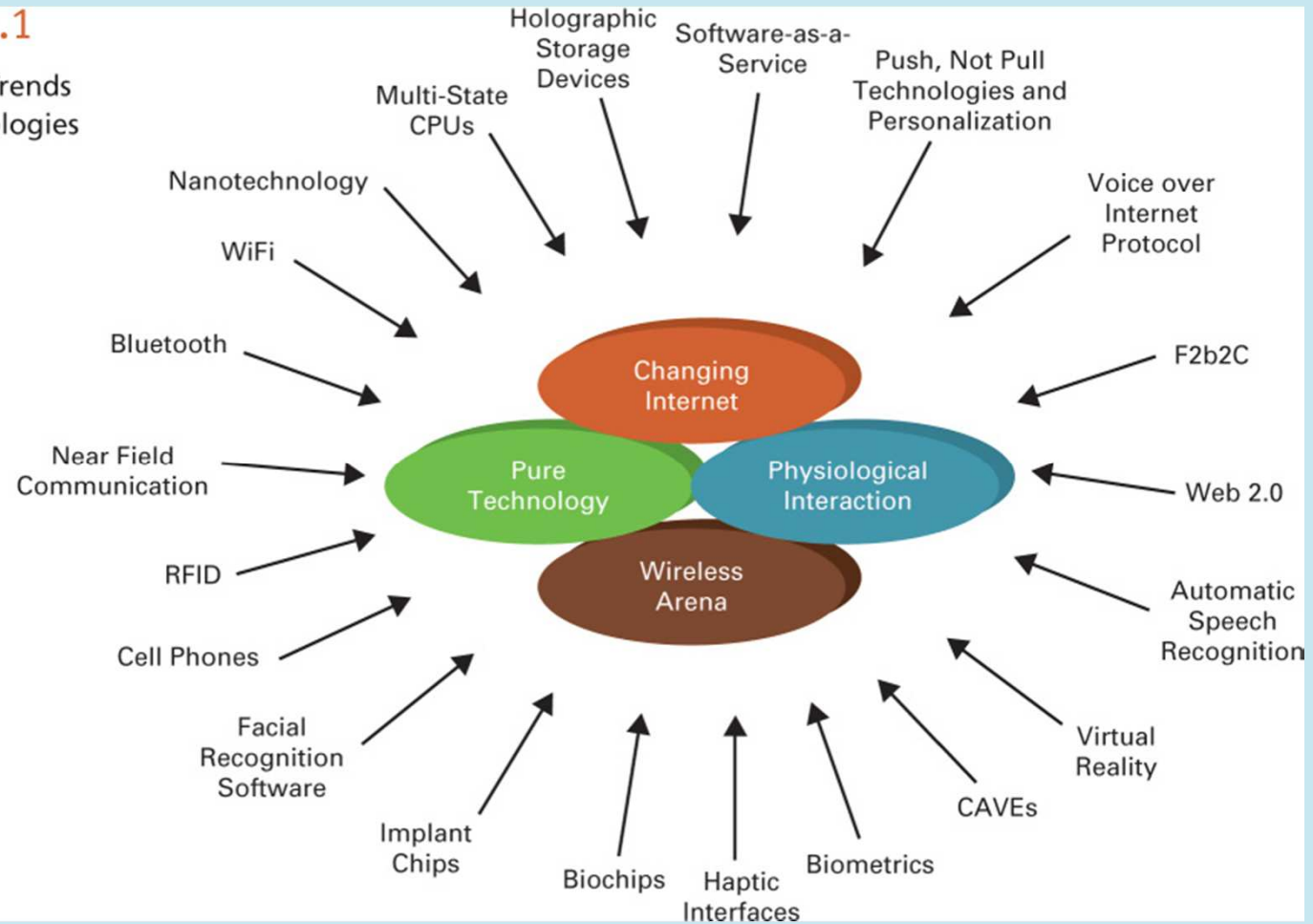
# INTRODUCTION

- **Technological changes will be unbelievable**
- **We need to focus on how they will change our personal and business lives**
- **Don't get caught up in only the technology itself**

# INTRODUCTION

Figure 9.1

Emerging Trends and Technologies



# THE CHANGING INTERNET

- **Software-as-a-service (SaaS)**
- **Push, not pull, technologies and personalization**
- **F2b2C**
- **Voice over Internet Protocol (VoIP)**
- **Web 2.0**

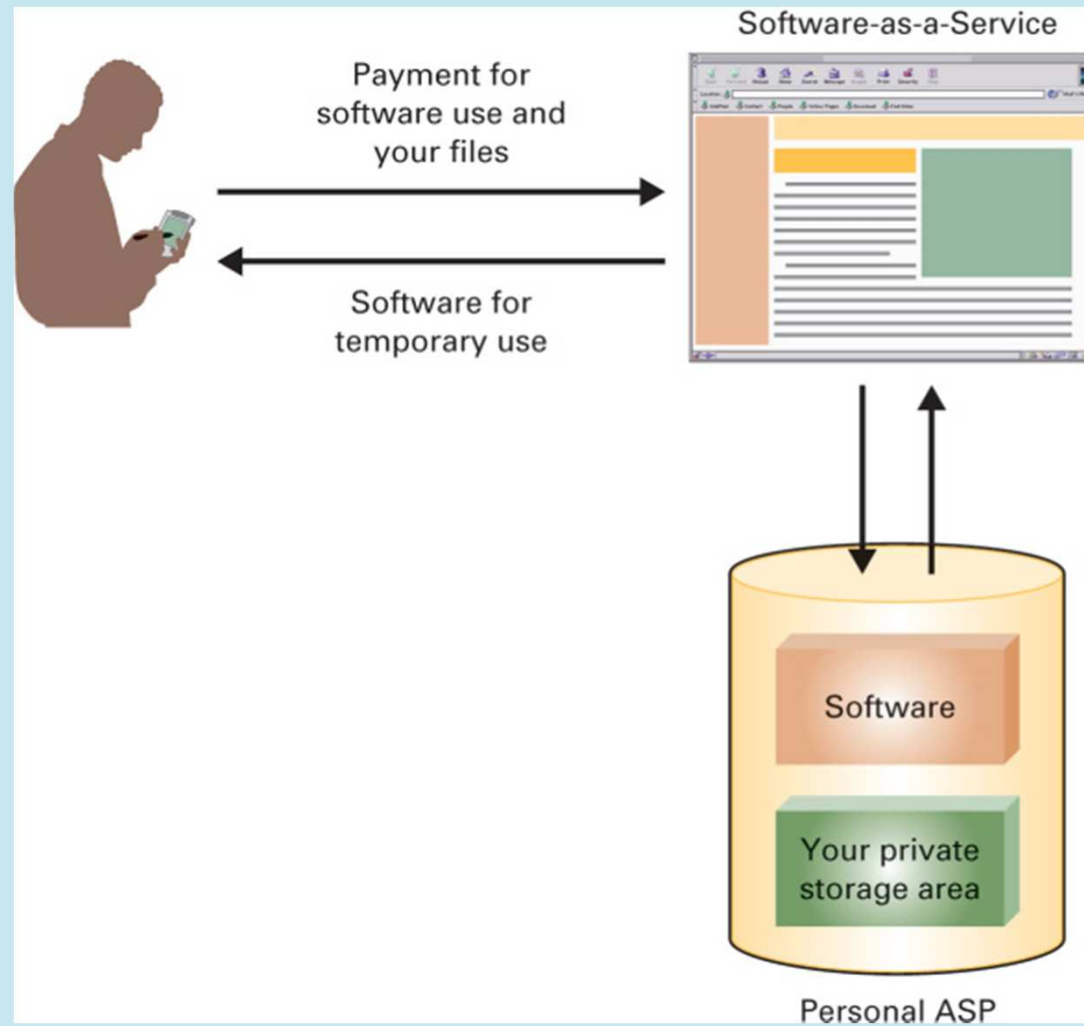
# Software-as-a-Service

- *Software-as-a-service (SaaS)*
  - **Delivery model for software in which you pay for software on a pay-per-use basis instead of buying it outright**
  - **Use any device anywhere to do anything**
  - **Pay a small fee and store files on the Web**
  - **Access those files later with your “regular” computer**
  - **Makes use of an application service provider**

# Software-as-a-Service

- *Application service provider (ASP)*
  - Supplies software applications (and other services such as maintenance, file storage, etc) over the Internet that would otherwise reside on customers' computers
  - Now, mainly limited to business applications
  - Future, personal ASPs renting software to you

# Software-as-a-Service



## Push vs. Pull Technologies and Personalization

- We live in a “pull” environment
  - Visit Web sites and request information, products, and services
- The future is a “push” environment
  - Businesses come to you with information, services, and product offerings
  - Based on your profile
  - Anticipating your interest

## Push vs. Pull Technologies and Personalization

- *Push* is **not** spam or mass e-mail
- Businesses will know so much about you that they can tailor and customize offerings
- For example:
  - Your favorite restaurant keeps track of your dinner orders recently
  - Their software determines that you have ordered fresh mahimahi several times
  - You have provided the restaurant with your cell number
  - The special today is blackened mahimahi
  - Their system will send you a text message announcing the daily special

# F2b2C

- New e-commerce business model
- *F2b2C – Factory-to-business-to-Consumer*
  - consumer communicates through a business on the Internet
  - directly provides product specifications to a factory
  - factory makes the customized and personalized product to the consumer's specifications and ships it directly to the consumer
- The business is really only an intermediary between the consumer and the factory
- A form of **disintermediation** (bypassing intermediaries like retail stores)

# Voice over Internet Protocol (VoIP)

- *Voice over Internet Protocol (VoIP)*
  - Send voice communications over the Internet – avoiding typical long distance charges
  - Catching on quickly in the business world
  - A little slower in the personal world
  - Not everyone has high-speed in-home Internet access

# Web 2.0

- *Web 2.0*
  - 2<sup>nd</sup> generation of the Web with online collaboration
  - Users are not just viewers of content
  - Users also function as both creators and modifiers of
    - content, dynamic and customized information feeds, and more
  - Wikis
  - Social networking sites
  - Blogs
  - RSS feeds
  - Podcasting

# Web 2.0

- *Wiki*
  - Allows visitor to create, edit, change, and often eliminate content
  - Closer connection with customers by
  - Provide enabling technologies that allow people – instead of paid employees – to create, modify, and oversee the development of products/services
- *Social networking site*
  - sites like Myspace, Facebook, etc where you post information about yourself, find friends, and so on

# Web 2.0

- *Blog*
  - Web site in the form of a journal
  - Owner posts entries and viewers can provide comments / responses
- *RSS feed*
  - Provides frequently published and updated digital content on the Web
  - Subscribing brings updated material automatically
- *Podcasting / videocasting*
  - Ability to download audio / video files for viewing and listening to on portable devices and computers

# PHYSIOLOGICAL INTERACTION

- Now, you use keyboards, mice, etc.
- These are physical interfaces
- Physiological interfaces will capture and use your real body characteristics
  - Voice
  - Iris scan
  - Fingerprints
  - Etc.

# Automatic Speech Recognition

- *Automatic speech recognition (ASR)*
  - Not only captures spoken words but also distinguishes word groupings to form sentences
  - Becoming more a reality everyday
  - Commercial systems cost less than \$100
  - Available in Microsoft Office
  - *Using the Microsoft Office automatic speech recognition tool I created this entire sentence. It took proximately ten seconds to create it, including formatting in italics.*

## ASR 3 Step Process

1. *Feature analysis* – captures words and converts them into phonemes (syllables)
2. *Pattern classification* – matches phonemes to words in an acoustic model database
3. *Language processing* – makes sense of what you're saying by choosing the best words

# Virtual Reality

- *Virtual reality*
  - Three-dimensional computer simulation in which you actively and physically participate
- Uses 3 unique devices
  - Glove
  - Headset
  - Walker

# Virtual Reality Devices

- *Glove* – input device; captures movement and strength of your hands and fingers
- *Headset (head-mounted display)* – I/O device; captures your head movement; screen covers your field of vision
- *Walker* – input device; captures movement of your feet as you walk or turn

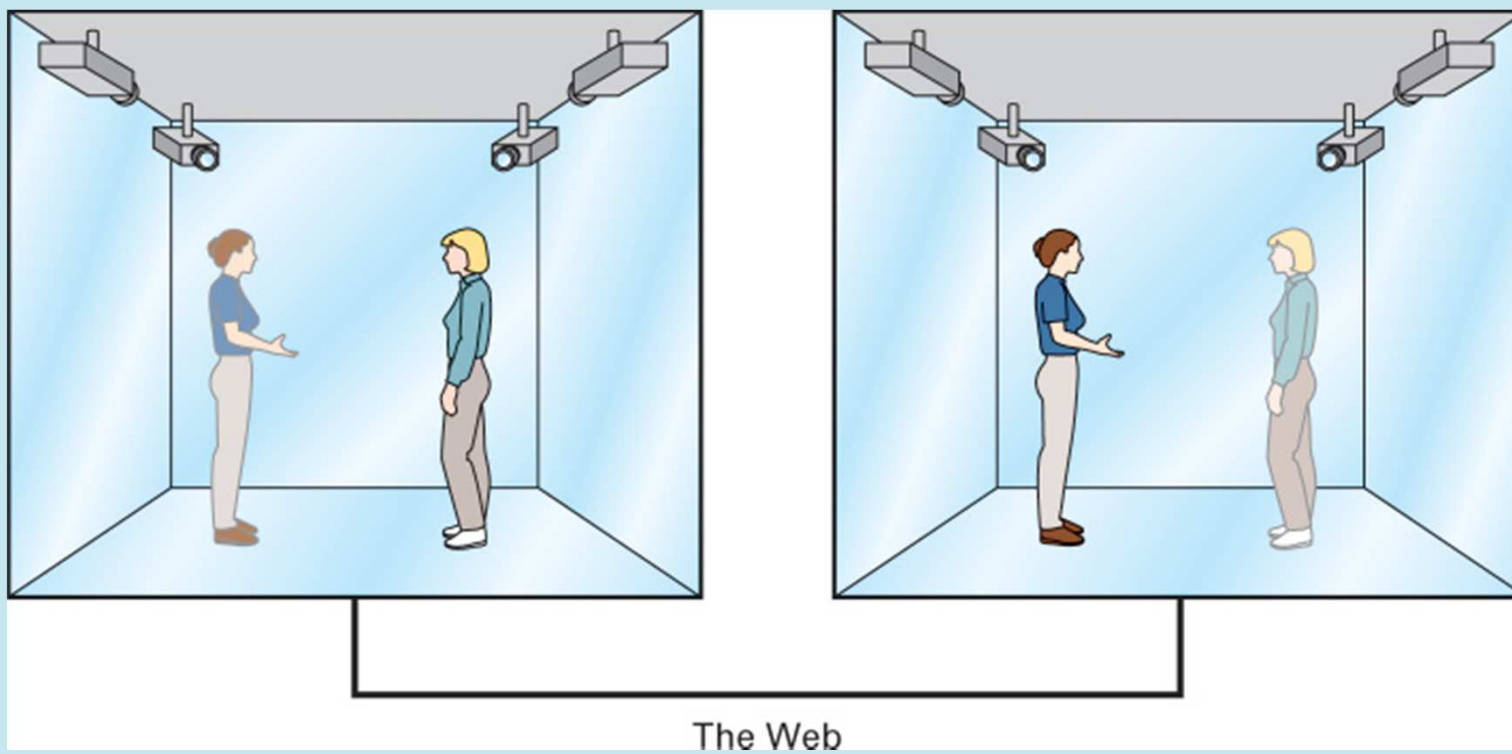
# Virtual Reality Applications

- Matsushita – design your own virtual kitchen
- Volvo – demonstrate car safety features
- Airlines – train pilots for adverse weather conditions
- Motorola – train assembly line workers
- Health care – train doctors in surgery on virtual cadavers

# Cave Automatic Virtual Environment

- *Cave automatic virtual environment (CAVE)* – special 3-D virtual reality room that can display images of people and objects in other CAVEs
- These are holographic devices
- *Holographic device* – creates, captures, and/or displays images in 3-D form

# Cave Automatic Virtual Environment



# Cave Automatic Virtual Environment

- Visit friends and family without getting on an airplane
- Customer service – the agent will appear next to you when you make a call
- The possibilities are limitless

# Haptic Interfaces

- *Haptic interface* – technology to add the sense of touch to an environment that previously only had visual and textual elements
  - Stationary snowboard game in which the snowboard moves and rocks
  - Game controllers vibrate when you crash your car
  - Wii, which has revolutionized the home video game market

# Biometrics

- *Biometrics* – the use of physiological characteristics – fingerprint, iris, voice sound, and even breath – to provide identification
- That's the narrow definition
- Can also create custom-fitting clothes using biometrics

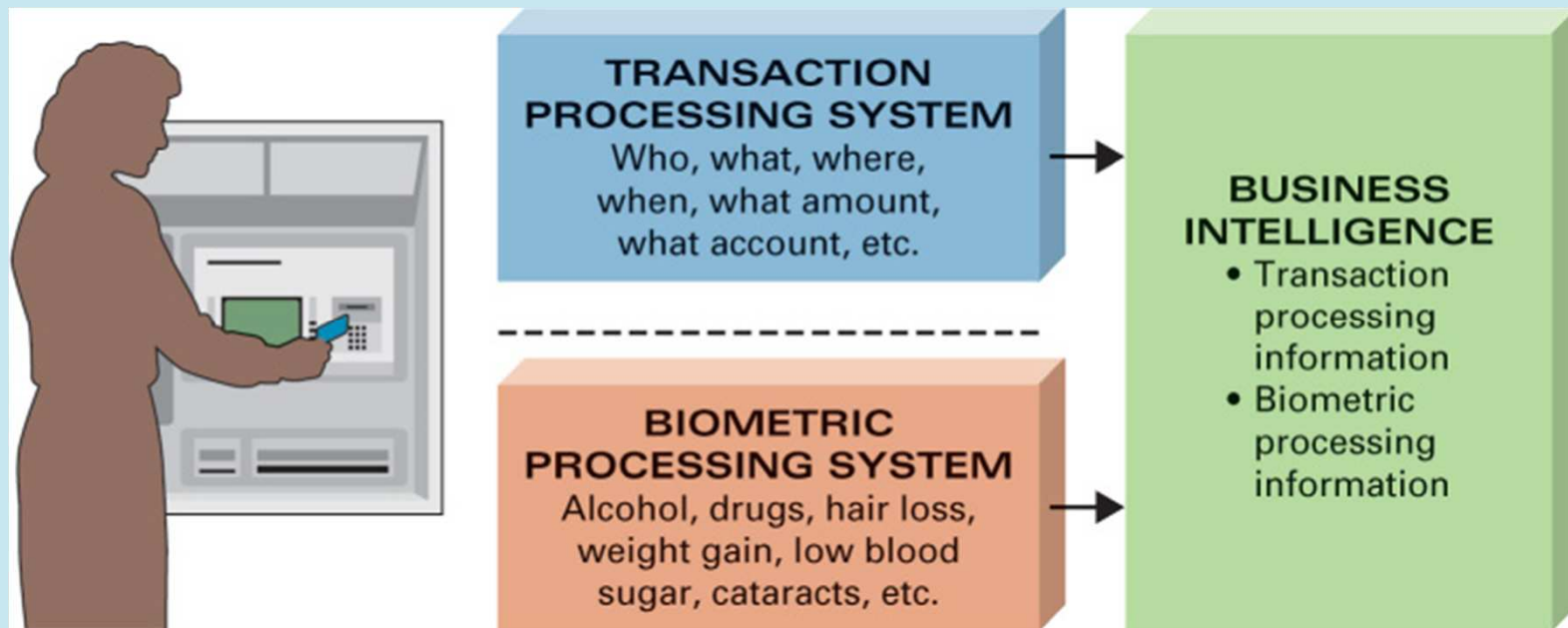
# Biometric Security

- **Best security is 3-step**
  1. **What you know (password)**
  2. **What you have (card of some sort)**
  3. **Who you are (biometric)**
- **Today's systems (ATMs for example) use only the first two**
- **One reason why identity theft is so high**

## Integrating Biometrics with Transaction Processing

- **TPS – captures events of a transaction**
- **Biometric processing system – captures information about you, perhaps...**
  - **Weight loss**
  - **Pregnancy**
  - **Use of drugs**
  - **Alcohol level**
  - **Vitamin deficiencies**

# Integrating Biometrics with Transaction Processing



## Integrating Biometrics with Transaction Processing

- Is this ethical?
- Can banks use ATMs and determine if you've been drinking?
- How will businesses of the future use biometric information?
  - Ethically?
  - Or otherwise?

## Other Biometric Devices

- *Biochip* – chip that can perform physiological functions when inserted into the human body
- *Implant chip* – microchip implanted into the human body that stores information about you and can be used for tracking (GPS)
  - Family of 4 in Florida already have them
- *Facial recognition software* – provides identification by evaluating facial characteristics

# THE WIRELESS ARENA

- Provides tremendous mobility
- Will dramatically change everything
- *Bluetooth* – short-range communication of about 30 feet
- *WiFi* – longer-range communication of up to about several miles
- The next generation of cell phone technology
- RFID

# Next Generation Cell Phones

- **Hard disk (upwards of 2Gb now)**
- **Processor capability**
- **Music enhancements**
- **Video support**
- **Apple's iPhone**

## Next Generation Cell Phones

- *Mobisodes* – short one-minute video clips of TV shows designed for viewing on a small cell phone screen
  - Download periodically
  - Combine to watch the entire show
  - Watch for them, these are coming soon (to a cell phone near you)

# Next Generation Cell Phones

- There is a downside
- Cell phones = next great playground for hackers and virus attacks
- No good anti-virus software for cell phones right now
- Don't download ring tones from Web sites – you could be downloading a virus

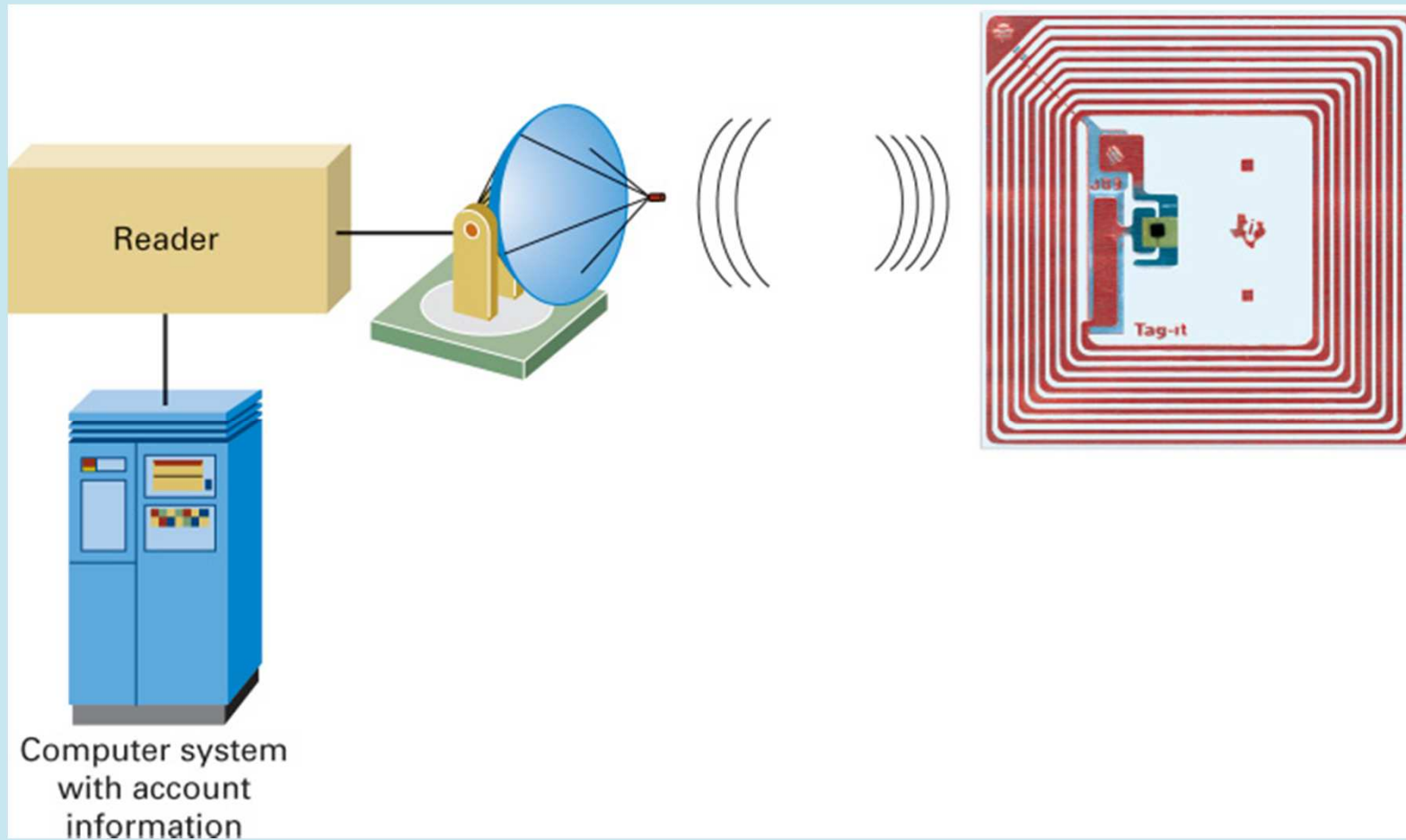
# RFID

- *RFID (radio frequency identification)*
  - uses a chip in a tag or label to store information
  - information is transmitted from, or written to, the tag or label when the chip is exposed to the correct frequency of radio waves
  - Wal-Mart is always in the business news about its requirement that all suppliers use RFID on products

# RFID

- **Common RFID is passive**
- **No battery power**
- **Antenna absorbs radio waves and stores as energy**
- **When enough energy is stored, the chip is “jolted” to life and information transmissions occur**

# RFID



# RFID Future

- Each and every product will have an RFID (EPC)
- Uniquely identifies each product
  - Expiration date
  - Places traveled
  - Etc

# PURE TECHNOLOGIES

- **Many will have broad applications in numerous areas**
  - **Nanotechnology**
  - **Multistate CPUs**
  - **Holographic storage devices**

# Nanotechnology

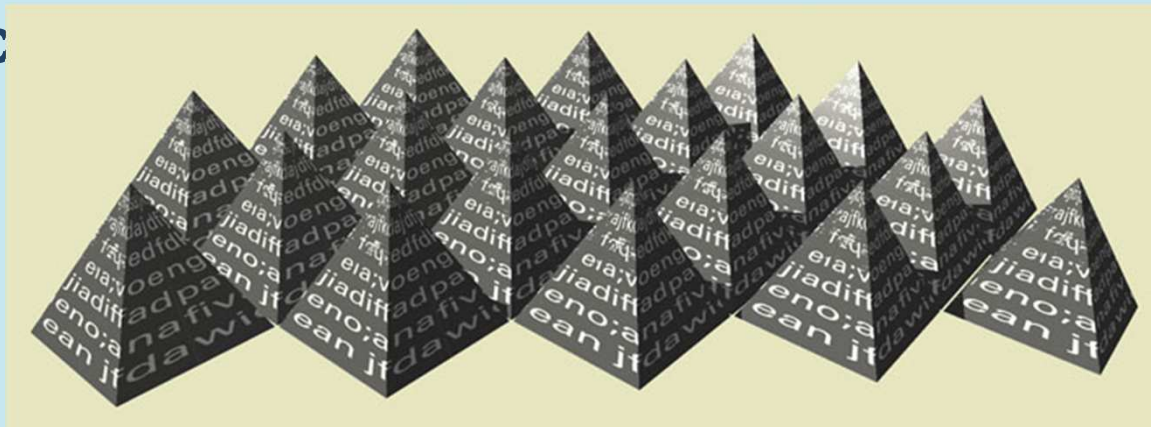
- *Nanotechnology*
  - control of matter at the atomic and sub-atomic levels
  - used to build devices on the same small scale
- Current approach – start big and squeeze, press, slice, and dice to make things small
- Nanotechnology approach – start with the smallest element possible (i.e., atom) and build up

## Multi-State CPUs

- Right now, CPUs are binary-state (0 and 1)
- *Multi-state CPU* – works with information represented in more than just 2 states, probably 10 states with each state representing a number from 0 through 9
- This will make small computers very fast

# Holographic Storage Devices

- *Holographic storage device* – stores information on a storage medium that is composed of 3-D crystal-like objects with many sides or faces
- This will provide tremendous storage in a small space



## **MOST IMPORTANT CONSIDERATIONS**

- **The necessity of technology**
- **Closing the great digital divide**
- **Technology for the betterment of society**
- **Exchanging privacy for convenience**
- **Ethics, ethics, ethics**

# The Necessity of Technology

- It's everywhere
- It's inescapable
- It's up to you how it gets used

# Closing the Great Digital Divide

- **There are technology-challenged countries and cultures**
- **You must take technology to those places**

## Technology for the Betterment of Society

- **Businesses use IT to make money and that's okay**
- **Technology can also be used when no money is to be made; and that's great**
- **Many medical applications of IT will never make any money**

# Exchanging Privacy for Convenience

- We do this everyday – in small ways
- Be careful
- We need to be aware of what privacy we give up

# **Ethics, Ethics, Ethics**

- **Ethics are essential**
- **Our society cannot operate without them**
- **Use technology wisely**
  - **For financial gain**
  - **But never to the detriment of other people**